Assignment 5

Subject - CCL
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Batch - B3

Problem Statement

Create a new Docker image such as Apache Tomcat, Python, Flask, or Node.js without using existing application base images (like node, python official images).

Code -

1] app.js

```
const http = require('http');

const server = http.createServer((req, res) => {
  res.writeHead(200, {'Content-Type': 'text/plain'});
  res.end('Hello from custom Docker image!');
});

server.listen(3000, () => {
  console.log('Server running on port 3000');
});
```

2] Dockerfile

```
FROM ubuntu:22.04

RUN apt-get update && apt-get install -y \
    curl \
    ca-certificates

RUN curl -fsSL https://deb.nodesource.com/setup_18.x | bash - \
    && apt-get install -y nodejs
```

WORKDIR /app

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]

Step 1: Build Docker Image

```
C:\Users\sansk\docker-assignment>docker build -t my-node-app .
[+] Building 193.7s (11/11) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                     0.1s
 => => transferring dockerfile: 476B                                                                                     0.1s
 => [internal] load metadata for docker.io/library/ubuntu:22.04                                                          4.2s
 => [auth] library/ubuntu:pull token for registry-1.docker.io                                                            0.0s
 => [internal] load .dockerignore                                                                                        0.1s
 => => transferring context: 2B                                                                                          0.1s
 => [1/5] FROM docker.io/library/ubuntu:22.04@sha256:eb29ed27b0821dca09c2e28b39135e185fc1302036427d5f4d70a41ce8f  14.8s
 => => resolve docker.io/library/ubuntu:22.04@sha256:eb29ed27b0821dca09c2e28b39135e185fc1302036427d5f4d70a41ce8fd  0.0s
 => => sha256:de47083ed7d7e66ba5116fed0a5b036b7c75ac74b2cfb0d9c3b89c79371c4a17 29.74MB / 29.74MB                   13.4s
 => => extracting sha256:de47083ed7d7e66ba5116fed0a5b036b7c75ac74b2cfb0d9c3b89c79371c4a17                            1.2s
 => [internal] load build context                                                                                        0.1s
 => => transferring context: 306B                                                                                        0.1s
 => [2/5] RUN apt-get update && apt-get install -y     curl     ca-certificates                                     81.0s
 => [3/5] RUN curl -fsSL https://deb.nodesource.com/setup_18.x | bash -     && apt-get install -y nodejs             79.2s
 => [4/5] WORKDIR /app                                                                                                   0.1s
 => [5/5] COPY app.js .                                                                                                  0.0s
 => exporting to image                                                                                                  13.8s
 => => exporting layers                                                                                                 10.0s
 => => exporting manifest sha256:58523bd1af96bbac4d04a8f29d2f758459c99660698f2fa6fc099654f6a2a601                    0.0s
 => => exporting config sha256:7312e007aa132ef9127e3d33e2cab0d1b3892c11de1e955e3be1959bf9957e83                        0.0s
 => => exporting attestation manifest sha256:3b014b7a08b4c4ac9255ccb2275365335eedab7ec30980f6f8f3039b7c85f7e1          0.0s
 => => exporting manifest list sha256:c43cd19f46fa9d4d662a6129c35430b3c4bb7fc196d9a47c1d8a884e87bc497d                 0.0s
 => => naming to docker.io/library/my-node-app:latest                                                                    0.0s
 => => unpacking to docker.io/library/my-node-app:latest                                                                 3.6s
```

Step 2: Run Container

```
C:\Users\sansk\docker-assignment>docker run -d -p 3000:3000 my-node-app
ef94cbd2412887a2c644986465b9b619216fb84dfeb6917954f3107ff0bd45e9
```

Step 3: Check running containers

```
C:\Users\sansk\docker-assignment>docker ps
CONTAINER ID   IMAGE         COMMAND         CREATED         STATUS         PORTS                                       NAMES
ef94cbd24128   my-node-app   "node app.js"   9 seconds ago   Up 8 seconds   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   blissful_curran
```

Step 4: Login to Docker Hub

```
C:\Users\sansk\docker-assignment>docker login
Authenticating with existing credentials... [Username: sanskruti316]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

Step 5: Tag Image

```
C:\Users\sansk\docker-assignment>docker tag my-node-app sanskruti316/my-node-app

C:\Users\sansk\docker-assignment>docker push sanskruti316/my-node-app
Using default tag: latest
The push refers to repository [docker.io/sanskruti316/my-node-app]
27e23732ccdf: Pushed
c9ca875aee9c: Pushed
55235c0e258b: Pushed
de47083ed7d7: Pushed
da8527626411: Pushed
5f6e9c58acc2: Pushed
latest: digest: sha256:c43cd19f46fa9d4d662a6129c35430b3c4bb7fc196d9a47c1d8a884e87bc497d size: 856
```

Step 6: Push Image

```
C:\Users\sansk\docker-assignment>docker tag my-node-app sanskruti316/my-node-app

C:\Users\sansk\docker-assignment>docker push sanskruti316/my-node-app
Using default tag: latest
The push refers to repository [docker.io/sanskruti316/my-node-app]
27e23732ccdf: Pushed
c9ca875aee9c: Pushed
55235c0e258b: Pushed
de47083ed7d7: Pushed
da8527626411: Pushed
5f6e9c58acc2: Pushed
latest: digest: sha256:c43cd19f46fa9d4d662a6129c35430b3c4bb7fc196d9a47c1d8a884e87bc497d size: 856
```

Step 7: Pull Image (Verification)

```
C:\Users\sansk\docker-assignment>docker pull sanskruti316/my-node-app
Using default tag: latest
latest: Pulling from sanskruti316/my-node-app
Digest: sha256:c43cd19f46fa9d4d662a6129c35430b3c4bb7fc196d9a47c1d8a884e87bc497d
Status: Image is up to date for sanskruti316/my-node-app:latest
docker.io/sanskruti316/my-node-app:latest
```

Step 8: Run Pulled Image

```
C:\Users\sansk\docker-assignment>docker stop ef94cbd24128
ef94cbd24128

C:\Users\sansk\docker-assignment>docker run -p 3000:3000 sanskruti316/my-node-app
Server running on port 3000
```

---

5. Output

Repositories

All repositories within the **sanskruti316** namespace.

Search by repository name | All content | Create a repository

Name	Last Pushed	Contains	Visibility	Scout
sanskruti316/my-node-app	6 minutes ago	IMAGE	Public	Inactive

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```
Hello from custom Docker image!
```